

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
LABORATORY OBSERVATION

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Branch & Sec: CSM-B

Date of Execution: 02 / 08 / 2026

Course Name: Full Stack Web Development

Course Code: 23CM4121

EXPERIMENT NO: 2

1. TITLE OF THE EXPERIMENT:

Implementation of HTML5 Canvas Graphics and CSS3 Styling Concepts

2. MAPPED OUTCOMES:

- **Course Outcome (CO):** CO1
- **Program Outcomes (POs):** PO1, PO2, PO3, PO4, PO5

3. AIM:

To design and implement a modern, responsive webpage titled "My Styled Webpage" demonstrating CSS3 styling properties and HTML5 Canvas.

4. ALGORITHM / DESIGN:

Algorithm: Design of My Styled Webpage

Step 1: Document Structure & CSS3 Reset — Initialize HTML5 template with `<meta name="viewport">` and set up global reset (margin: 0, padding: 0, box-sizing: border-box). Apply a linear gradient background and Flexbox layout to `<body>`.

Step 2: Header & Navigation Bar — Create `<header>` with text shadows and deep background color. Add `<nav>` using horizontal Flexbox layout with anchor links featuring smooth CSS hover background-color transitions.

Step 3: HTML5 Canvas Section (Part A) — Add a `<canvas>` element inside a centered container card. Use JavaScript (`getContext('2d')`) to draw a stroked rectangle (`fillRect`, `strokeRect`), a colored circular arc (`arc`, `fill`), an angled line (`moveTo`, `lineTo`), and custom centered text (`fillText`).

Step 4: Responsive Cards Grid (Part B) — Implement a CSS Grid container (`repeat(auto-fit, minmax(280px, 1fr))`) to display three content `<article>` cards. Apply border-radius, box-shadow, and smooth hover elevation effects (`transform: translateY`) to cards and interactive buttons.

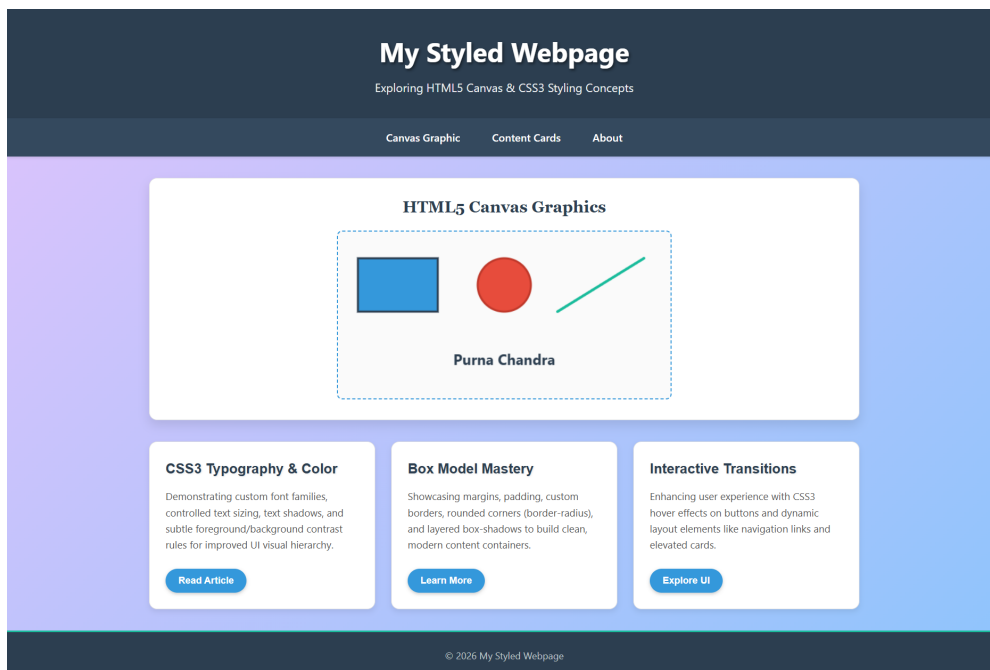
Step 5: Footer & Closing — Add a <footer> with copyright information and a styled top border, then close all HTML tags.

5. TEST CASES & EXECUTION DATA:

Test Cases (Validation & UI Verification)

- **TC-01:** Input: Load the webpage in a modern web browser
Expected: Background displays a smooth diagonal linear gradient and header renders with drop shadows | Status: Passed
 - **TC-02:** Input: Hover cursor over navigation bar links
Expected: Background smoothly transitions to #1abc9c with white text | Status: Passed
 - **TC-03:** Input: Execute JavaScript Canvas API rendering script
Expected: Canvas displays a blue stroked rectangle, red filled circle, green angled line, and centered text ("Purna Chandra") | Status: Passed
 - **TC-04:** Input: Hover cursor over individual content cards
Expected: Card elevates vertically (translateY(-5px)) with expanded shadow depth | Status: Passed
 - **TC-05:** Input: Hover cursor over ".btn" action buttons
Expected: Button background darkens and scales smoothly | Status: Passed
 - **TC-06:** Input: Resize browser viewport across different screen widths
Expected: CSS Grid automatically reflows content cards into single or multi-column layouts responsively | Status: Passed
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6. OUTPUT PROOF:



Github Link:

https://github.com/purnachandra1052006-collab/FullStack/blob/main/Observation/Task_2/My_Styled_Webpage.html

7. LEARNING OUTCOME / RESULT: Successfully implemented and verified the styling and graphics script. Learned how to enhance static web pages with modern CSS3 visual techniques (CSS Grid, Flexbox, transitions, pseudo-classes, gradients) and dynamically render 2D geometric shapes and text using the HTML5 <canvas> API and JavaScript context.

RUBRIC EVALUATION (Faculty Use Only)

1. **Design & Logic** (5 Marks):

2. **Execution & Output** (5 Marks):

3. **Viva & Record** (5 Marks):

TOTAL MARKS: [/ 15]

Faculty Signature: _____